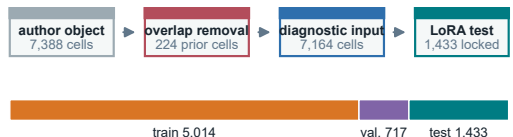
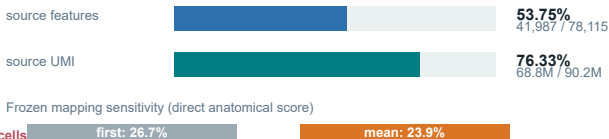


a A provenance gate prevents reuse of prior strict-transfer cells
GSE270342 wheat-root author object; labels remain unseen by frozen inference



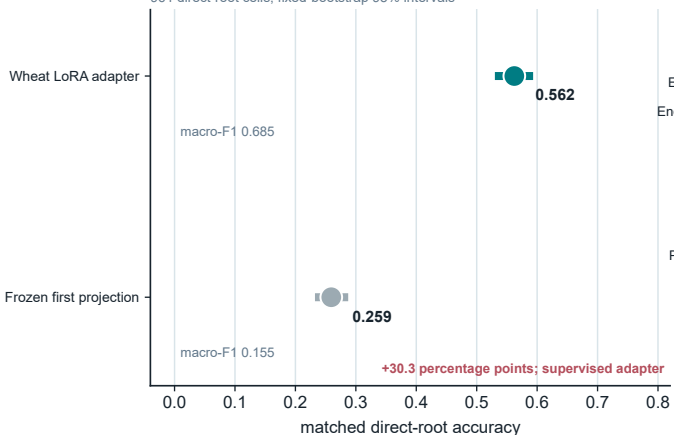
Same study remains explicit; barcode removal prevents reuse of the recorded strict-transfer cells

b An author orthogroup contract preserves most counted expression
Many-to-many groups are recorded; frozen inference uses a deterministic first target



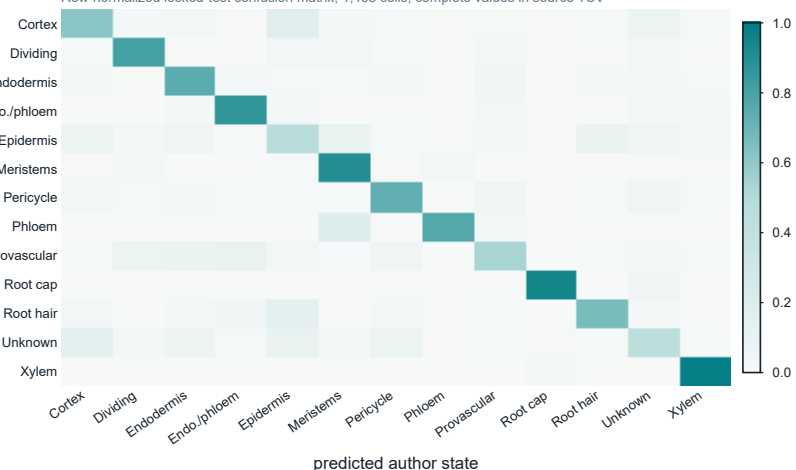
c Labelled wheat adaptation improves matched direct-root states

964 direct-root cells; fixed-bootstrap 95% intervals



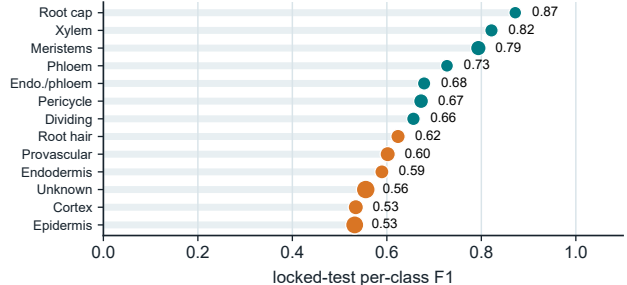
d The adapted head resolves all 13 author labels

Row-normalized locked-test confusion matrix; 1,433 cells; complete values in source TSV



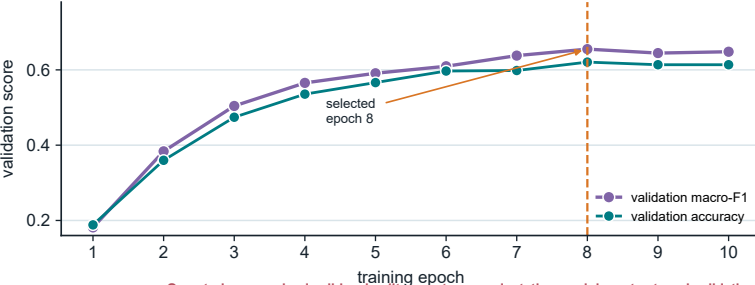
e Rare and mixed root states remain visible

Point area scales with class support; no author classes are hidden



f Selection is isolated from locked testing

Best validation epoch: 8; 13-class locked test: 62.25% accuracy and 0.6660 macro-F1



One study, supervised cell-level split; report as an adaptation module, not external validation.